METRIC

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

LEVEL, SURVEYING, SELF-LEVELING, AND TILTING

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE.

1.1 Abstract. This commercial item description covers levels (instruments) used for taking differential elevation measurements, in surveying and other engineering applications. The complete instrument shall consist of an internal focusing telescope capable of 360 degree rotation, image lens system, cross hairs (horizontal and vertical), leveling system, tripod, and accessories.

2. CLASSIFICATION.

The levels shall be classified as follows:

Type I	–	Tilting level
Style A	–	With plane-parallel optical micrometer attachment
Style B	–	Without plane-parallel optical micrometer attachment
Type II	–	Self-leveling level
Style A	–	With horizontal circle graduations
Style B	–	Without horizontal circle graduations
Class 1	–	1st order accuracy
Class 2	–	2nd order accuracy

3. SALIENT CHARACTERISTICS.

3.1 Accuracy. Accuracy shall be certifiable, for the life of the level, to permit not greater than 1.54 seconds of error (1st order) for class 1 instruments; and not greater than 3.07 seconds of error (2nd order) for class 2 instruments.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: U.S. Army Topographic Engineering Center, ATTN: CETEC-TD-T, 7701 Telegraph Road, Alexandria, VA 22315-3864.

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3.2 Level types.

3.2.1 Tilting level. A tilting instrument (type I) shall require manual leveling prior to use and periodic checking for maintenance of a level horizontal line of sight during use. The instrument shall be provided with a circular spirit level, or similar device, for initial coarse leveling and a graduated leveling readout device and leveling screws for precise leveling. The type I, style A level shall be equipped with a parallel-plane optical micrometer that allows the operator to measure horizontal distances, within instrument limits. The type I, style B level shall not be equipped with an optical micrometer.

3.2.2 Self-leveling. A self-leveling instrument (type II) shall automatically maintain the telescope in a level position, until no longer needed, once the instrument has been manually brought to a defined approximation of level. Type II instruments shall have provisions for measuring horizontal distances. The type II, style A instrument shall have provisions for horizontal full circle angle measurement; the style B instrument shall not. Type II levels shall be equipped with an automatic compensator, with a working range of not less than ± 15 seconds of arc, that shall maintain a true horizontal level line of sight at all telescope positions. After telescope rotation, the instrument shall return to true level in not greater than 5 seconds.

3.3 Optical system. The optical system of either type instrument shall be aberration free, provide a clear, bright, high contrast, and sharp image throughout the focusing range and field of view. The image may be erect or inverted. Optical properties shall be as follows:

a. Magnification: Not less than 30 diameters at an accuracy of ± 5 percent
b. Objective lens effective aperture: Not less than 39 millimeters (mm)
c. Minimum focus distance: Not greater than 3.7 meters
d. Minimum field of view: Not less than 60 minutes of arc
e. Resolution: Not less than 4 seconds of arc
f. Eye relief distance: Not less than 7 mm

3.3.1 Reticle. The reticle shall be a pattern of permanent opaque lines, on a distortion free, optically transparent medium. It shall be adjustable for precise positioning of the cross hairs, in the horizontal and vertical planes, about the central line of sight axis of the telescope. The reticle shall be positioned in the eyepiece focus and so as to eliminate distortion during collimation. Horizontal and vertical stadia lines shall be a minimum of 5 minutes of arc in length, parallel to the corresponding cross lines, and spaced at standard stadia line intervals of 1 to 100. The width of all reticle lines shall be 3 seconds of arc, + 0.0 seconds, -0.5 seconds.

3.3.2 Stadia constant. When horizontal distance is measured using the stadia method the difference shall not be greater than ± 0.3 percent when compared to the verified distance.

3.3.3 Optical micrometer. When provided, the parallel plate optical micrometer shall measure the interval between the horizontal reticle line and the nearest rod graduation directly to 1 mm.

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3.4 Level indicators.

3.4.1 Coarse level indicator. The coarse leveling indicators shall be clearly visible when the instrument is set up, horizontal, and parallel to the line of sight. The level indicator sensitivity shall be by instrument type as follows:

a. Type I 10 ± 2 minutes of arc per 2 mm division.
b. Type II 7 to 15 minutes of arc per 2 mm division.

3.4.2 Telescope (precise) level indicator. The type I telescope level indicator shall have a sensitivity of 8 to 10 seconds of arc, per 2 mm movement. Level indicator readings shall be clearly discernible in temperatures of -25° to +120°F.

3.5 Mounting. The instrument shall be mounted on a tribrach with a baseplate and leveling screws. Leveling screws shall operate without binding, backlash, or end play in temperatures of -25° to +120°F. For interface with existing instruments, the tribrach shall have at its center a means of accepting an instrument mounting screw with .625 (5/8)–11 UNC–2B thread in accordance with ASME B1.1. When leveled, the instrument shall smoothly and easily rotate through a full horizontal circle without binding or end play.

3.6 Horizontal circle. When provided, the horizontal circle shall have index marks graduated in one-degree intervals, labeled at 10-degree intervals, in a clockwise direction. Angle readings to the nearest 10 minutes shall be visible through a supplied magnifying viewer sufficient to estimate the reading to the nearest minute. The circle shall have a fine adjustment capability for precise positioning of the vertical cross hair in the horizontal circle plane.

3.7 Tripod. The level shall be provided with a wide frame tripod having collapsible legs. A means to secure the legs when collapsed in transport configuration shall be provided. The overall length of the tripod, with legs secured but fully extended, shall not be less than 168 cm (66 inches). A head or adapter ring to interface with the tribrach mounting threads shall be supplied. A thread protective cap to be used when the instrument is not mounted on the tripod shall be supplied. A means of secure storage for the cap, when not in use, shall be provided on the tripod. The tripod material shall be of sufficient strength to rigidly support the instrument and rugged enough to withstand large changes in temperature or humidity. Leg ends shall be configured for penetration of hard natural surfaces with the application of force.

3.8 Accessories. The instrument shall be provided with the following accessories.

a. Lens cleaning brush
b. Screwdriver for instrument adjustment as required
c. Adjusting pin for reticle adjustment as required
d. Objective lens dust cap
e. Eyepiece lens cover
f. Objective eyepiece sunshade
g. Waterproof cover for the instrument (level)

3.9 Carrying case. A carrying case for the instrument and accessories shall be provided. The case material shall be sufficiently rugged to withstand normal handling and usage. The case liner shall form fit the instrument and shall be securely attached to the case. The level shall be secured in the case. The case shall be provided with locking latches. Each case shall contain

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desiccant adequate for the volume of the case. Accessory items shall be compartmented in the case to prevent damage to themselves and the instrument during handling and transport. A means shall be provided for carrying the case and contents with the instrument in an upright position.

4. REGULATORY REQUIREMENTS.

4.1 Recycled, recovered, or environmentally preferable materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS.

5.1 Market acceptability. The levels offered shall be the latest version of the standard product of the supplier that has been used in the field for at least one year by commercial organizations.

5.2 Product conformance. The levels offered shall meet the salient characteristics of this CID, conform to the manufacture's drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING.

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order (see 7.1)

7. NOTES

7.1 Ordering data. Acquisition documents should include:

a. Title, number, and date of this document.
b. Type, style, and class of level required (see 2.1).
c. Level of preservation and packing required (see 6.1).

7.2 Intended use. The levels shall be used to determine differences in elevation between various pre-selected points. For type I, style A, and type II instruments the horizontal distance of the points from the instrument location may be determined. For type II style A instruments horizontal angular measurements between designated points may be determined.

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7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for CID A-A-52036A

AA52036XXX

- Class: Class 1 = First order; error not greater than 1.54 seconds
  Class 2 = Second order; error not greater than3.07 seconds
- Style: Style A = With plane parallel optical micrometer
  Style B = Without plane parallel optical micrometer
  Style A = With horizontal circle
  Style B = Without horizontal circle
- Type Type I = Tilting level
  Type II = Self leveling level
- Document identifier

Example: AA52036IID1—A self-leveling instrument, without a horizontal circle, capable of first order accuracy.

7.4 Sources of documents.

7.4.1 Copies of the Federal Acquisition Regulation (FAR) may be obtained from: Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.4.2 Copies of ASME B1.1 may be obtained from: American Society of Mechanical Engineers, United Engineering Center, 345 E. 47th St., New York, NY 10017.

7.5 Subject term (key word) listing.

Alidade
Collimation
Measurement
Optics
Transit
Tribrach
Tripod

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Custodian:
Army - CE3
Air force - 99
Navy – YD2
DLA - GS

Review activities:
Army-CR
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Civil Review Agency:
GSA - FCGC

Preparing activity:
Army - CE3

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